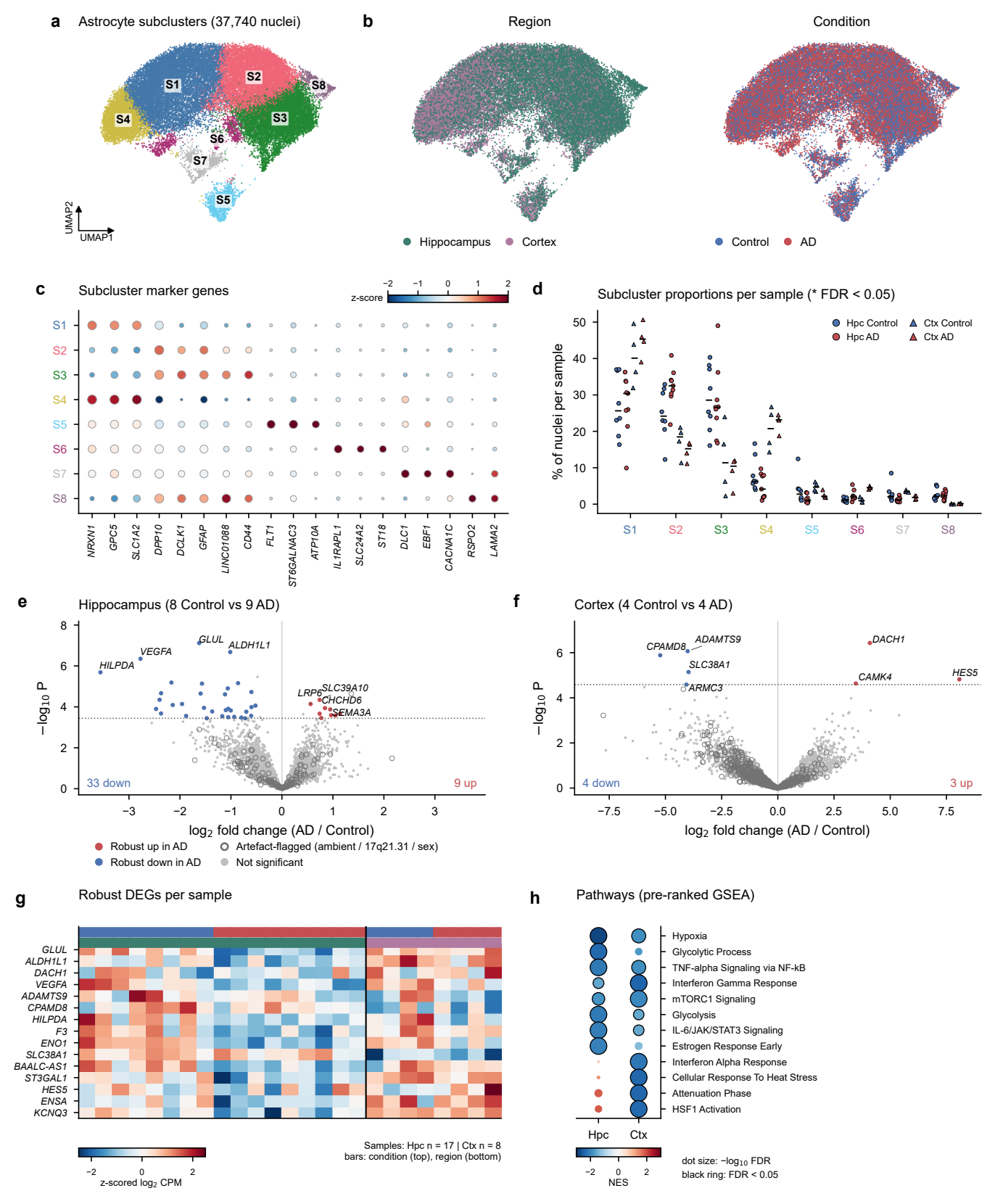

a
Astrocyte subclusters (37,740 nuclei)
S1
S2
S3
S4
S5
S6
S7
S8
UMAP2
UMAP1
b
Region
Hippocampus
Cortex
Condition
Control
AD
c
Subcluster marker genes
z-score
−2 −1 0 1 2
S1 S2 S3 S4 S5 S6 S7 S8
NRXN1 GPC5 SLC1A2 DPP10 DCLK1 GFAP LINC01088 CD44 FLT1 ST6GALNAC3 ATP10A IL1RAPL1 SLC24A2 ST18 DLC1 EBF1 CACNA1C RSPO2 LAMA2
d
Subcluster proportions per sample (* FDR < 0.05)
Hpc Control
Hpc AD
Ctx Control
Ctx AD
% of nuclei per sample
0 10 20 30 40 50
S1 S2 S3 S4 S5 S6 S7 S8
e
Hippocampus (8 Control vs 9 AD)
−log10 P
GLUL
ALDH1L1
VEGFA
HILPDA
LRP6
SLC39A10
CHCHD6
SEMA3A
33 down
9 up
log2 fold change (AD / Control)
Robust up in AD
Robust down in AD
Artefact-flagged (ambient / 17q21.31 / sex)
Not significant
f
Cortex (4 Control vs 4 AD)
−log10 P
CPAMD8
ADAMTS9
SLC38A1
ARMC3
DACH1
CAMK4
HES5
4 down
3 up
log2 fold change (AD / Control)
g
Robust DEGs per sample
GLUL
ALDH1L1
DACH1
VEGFA
ADAMTS9
CPAMD8
HILPDA
F3
ENO1
SLC38A1
BAALC-AS1
ST3GAL1
HES5
ENSA
KCNQ3
z-scored log2 CPM
Samples: Hpc n = 17 | Ctx n = 8
bars: condition (top), region (bottom)
h
Pathways (pre-ranked GSEA)
Hypoxia
Glycolytic Process
TNF-alpha Signaling via NF-kB
Interferon Gamma Response
mTORC1 Signaling
Glycolysis
IL-6/JAK/STAT3 Signaling
Estrogen Response Early
Interferon Alpha Response
Cellular Response To Heat Stress
Attenuation Phase
HSF1 Activation
Hpc
Ctx
NES
dot size: −log10 FDR
black ring: FDR < 0.05